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PlixLab: A Python package for browser-based interactive presentations

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Software

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Summary

Disseminating scientific results increasingly requires interactivity with figures and data. For example, enabling the rotation of a molecule within a presentation slide allows biologists to illustrate molecular docking mechanisms critical to drug discovery. In computational fluid dynamics, key insights are often encoded in spatial maps; allowing presenters to interactively explore these maps can lead to more engaging and effective communication. Despite these needs, mainstream presentation tools primarily support static content. `PlixLab` addresses this gap by combining the power of Python with modern JavaScript visualization libraries. `PlixLab` is a Python-based framework that generates JSON-encoded data, which is rendered interactively in the browser.

Statement of need

`PlixLab` is a Python library for creating browser-based interactive presentations. It supports a range of plugins, including interfaces to `3jmol` (Rego & Koes, 2015) for protein visualization, `plotlyjs` (Plotly, Inc, 2023) and `Bokeh` (Bokeh Development Team, 2018) for dynamic plots, and web embedding for integrating full scientific web applications directly into slides. Notably, when embedding a JupyterLite application (JupyterLite Contributors, 2021), `PlixLab` enables the use of an in-slide REPL or even a full `Jupyter Notebook` (Kluyver et al., 2016), with Python running locally in the browser via `Pyodide` (Pyodide development team, 2021). These capabilities make `PlixLab` a unified platform for building data-centric presentations, eliminating the need to juggle multiple tools. Standard presentation features, such as animations, can also be implemented programmatically.

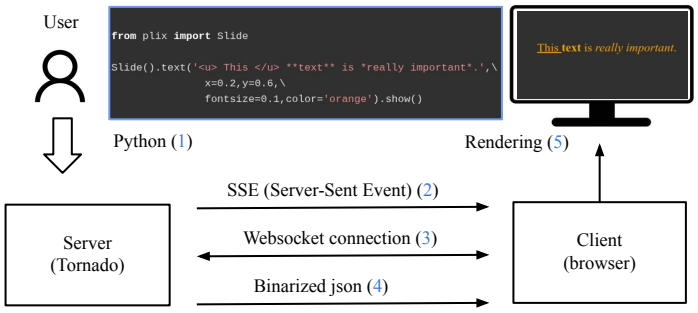

Figure 1: Real-time update flow in PlixLab. When a Python file is modified, the server sends an SSE (1–2) to the browser, which opens a WebSocket (3) to receive binarized JSON data (4) and re-render the presentation (5), enabling seamless hot-reloading..

The graphical user interface (GUI), written in JavaScript, offers three navigation modes:

single-slide view, grid view, and fullscreen presentation mode. To streamline slide development, `PlixLab` includes hot-reload functionality (see Figure 1). When a user modifies the Python source file, a local server is triggered, which uses Server-Sent Events (SSE) to notify the browser client. The client then opens a WebSocket connection to receive the updated binary JSON data and renders the new content. This design avoids inefficient client-side polling, resulting in a responsive and seamless development experience.

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